

Honors Analysis I

Recitation 4 Notes

GRAYSON DAVIS

Quiz 4 covering Homework 4 will be next week! Today I can either go through the midterm solutions or talk about metric spaces. These notes will cover the content on metric spaces. I'll write up solutions to the midterm separately.

1. CONCEPT REVIEW

A *metric space* is a tuple (X, d) of a set X and a *metric* $d: X \times X \rightarrow [0, \infty)$ which satisfies

- (1) $d(x, y) = 0$ if and only if $x = y$,
- (2) $d(x, y) = d(y, x)$ for all $x, y \in X$,
- (3) $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in X$.

An *open ball of radius $r > 0$ at $x \in X$* is the set

$$B(x, r) := \{y \in X : d(x, y) < r\}.$$

A *closed ball of radius $r > 0$ at $x \in X$* is the set

$$\overline{B}(x, r) := \{y \in X : d(x, y) \leq r\}.$$

A metric space is called *complete* if all Cauchy sequences converge. Let's do some examples:

Example 1.1. If d_1, d_2 are two given metrics on a set X , decide whether the following are metrics on X :

- (a) $d_1 + d_2$,
- (b) $\max(d_1, d_2)$.

Solution:

(a) Non-negativity is immediate from non-negativity of d_1 and d_2 . If $x = y$ then $d_1(x, x) + d_2(x, x) = 0$. Conversely, if $d_1(x, y) + d_2(x, y) = 0$, then $d_1(x, y) = -d_2(x, y) \leq 0$, so in fact $d_1(x, y) = 0$, hence $x = y$. Symmetry is also immediate from symmetry of d_1, d_2 . Given $x, y, z \in X$, we compute

$$\begin{aligned} d_1(x, y) + d_2(x, y) &\leq d_1(x, z) + d_1(z, y) + d_2(x, z) + d_2(z, y) \\ &= (d_1(x, z) + d_2(x, z)) + (d_1(z, y) + d_2(z, y)), \end{aligned}$$

so the triangle inequality holds. Thus, $d_1 + d_2$ is a metric.

(b) As in (a), non-negativity and symmetry are immediate. If $x = y$, then

$$\max(d_1(x, y), d_2(x, y)) = \max(0, 0) = 0.$$

If instead $\max(d_1(x, y), d_2(x, y)) = 0$, then $d_1(x, y) = 0$ or $d_2(x, y) = 0$. so that $x = y$. For the triangle inequality, given $x, y, z \in X$, we compute WLOG

$$\begin{aligned}\max(d_1(x, y), d_2(x, y)) &= d_1(x, y) \leq d_1(x, z) + d_1(z, y) \\ &\leq \max(d_1(x, z), d_2(x, z)) + \max(d_1(z, y), d_2(z, y)).\end{aligned}$$

Thus, $\max(d_1, d_2)$ is a metric. □

Example 1.2. Consider the metric space $(\mathbb{R}^n, d)$ where d is the usual Euclidean metric. Let e_i denote the i -th standard basis vector.

(a) Verify $d(e_i, e_j) = \sqrt{2}$ when $i \neq j$,

(b) Prove that each e_i is contained in exactly one ball in the collection

$$\mathcal{C} := \{B(e_i, \sqrt{2})\}_{i=1}^n \cup \{B(-e_i, \sqrt{2})\}_{i=1}^n,$$

(c) Prove that each $x \in B(0, 1)$ is contained in at least one of the balls in $\mathcal{C}$.

Solution:

(a) If $i \neq j$,

$$(e_i - e_j)_k = \begin{cases} 1 & k = i \\ -1 & k = j \\ 0 & \text{otherwise} \end{cases}.$$

Thus,

$$d(e_i, e_j) = |e_i - e_j| = \sqrt{\sum_{k=1}^n (e_i - e_j)_k^2} = \sqrt{2}.$$

(b) Since $d(e_i, e_i) = 0$, we immediately have $e_i \in B(e_i, \sqrt{2})$. It remains to check that e_i is in no other ball in the collection $\mathcal{C}$. If $i \neq j$, then $d(e_i, e_j) = \sqrt{2}$ by (a), and hence $e_i \notin B(e_j, \sqrt{2})$. Likewise a similar computation as in (a) shows that $d(e_i, -e_j) = \sqrt{2}$, so that $e_i \notin B(-e_j, \sqrt{2})$. Finally, observe that

$$d(e_i, -e_i) = \sqrt{(1 - (-1))^2} = 2 > \sqrt{2},$$

so that $e_i \notin B(-e_i, \sqrt{2})$.

(c) Suppose there is $x \in B(0, 1)$ that is contained in no ball of $\mathcal{C}$. Then

$$d(x, e_i), d(x, -e_i) \geq \sqrt{2}$$

for each $i = 1, \dots, n$. Then, in particular, we have

$$2 \leq d(x, e_1)^2 = |x - e_1|^2 = |x|^2 - 2|x_1| + 1.$$

Therefore, $1 > |x|^2 \geq 1 + 2|x_1| \geq 1$, a contradiction. Hence, x is contained in some ball in $\mathcal{C}$. □

Example 1.3. Let $X = \ell^1$ denote the set of real sequences $\{a_n\}_n$ for which $\sum_{n=1}^{\infty} |a_n| < \infty$. Let d_1 and d_∞ be metrics on X given by

$$d_1(x, y) = \sum_{n=1}^{\infty} |x_n - y_n|, \quad d_\infty(x, y) = \sup_{n \in \mathbb{N}} |x_n - y_n|$$

for every $x = \{x_n\}_n, y = \{y_n\}_n \in X$.

(a) Confirm these are both metrics.

(b) Show there are sequences $\{x^{(k)}\}_k$ in X which converge in (X, d_∞) but fail to converge in (X, d_1) .

(c) Show that if $\{x^{(k)}\}_k$ converges in (X, d_1) , then it automatically converges in (X, d_∞) .

Solution:

(a) This is left as an exercise.

(b) Define a sequence $\{x_k\}_k \subseteq X$ by

$$x_n^{(k)} := \begin{cases} 1/k & n \leq k \\ 0 & n > k, \end{cases}$$

i.e.,

$$x^{(k)} = (\underbrace{1/k, 1/k, \dots, 1/k}_{k \text{ times}}, 0, \dots).$$

Then

$$\sum_{n=1}^{\infty} |x_n^{(k)}| = \underbrace{1/k + \dots + 1/k}_{k \text{ times}} = 1 < \infty,$$

so that $x^{(k)} \in X$ for every $k \in \mathbb{N}$. We claim that $x^{(k)} \rightarrow 0$ in (X, d_∞) . Indeed,

$$\sup_{n \in \mathbb{N}} |x_n^{(k)}| = 1/k \rightarrow 0 \text{ as } k \rightarrow \infty.$$

To show non-convergence in (X, d_1) , suppose instead $x^{(k)} \rightarrow x$ as $k \rightarrow \infty$. If $x \neq 0$, then there is $m \in \mathbb{N}$ such that $x_m \neq 0$. But then

$$|x_m - x_m^{(k)}| \leq \sum_{n=1}^{\infty} |x_n - x_n^{(k)}| \rightarrow 0 \text{ as } k \rightarrow \infty,$$

a contradiction, since $x_m^{(k)} \rightarrow 0$ as $k \rightarrow \infty$. Thus, $x = 0$, which contradicts

$$\sum_{n=1}^{\infty} |x_n^{(k)}| = 1$$

for all $k \in \mathbb{N}$.

(c) Suppose $x^{(k)} \rightarrow x$ in (X, d_1) . Then

$$|x_n^{(k)} - x_n| \leq \sum_{n=1}^{\infty} |x_n^{(k)} - x_n|$$

Taking the supremum over all $n \in \mathbb{N}$ on the LHS and then taking the limit $k \rightarrow \infty$ concludes the proof.

□

Example 1.4. *Show that the metric space (X, d) is complete if and only if the metric space $(X, \tilde{d})$ is complete, where $\tilde{d} = d/(1 + d)$.*

Solution:

Assume first that (X, d) is complete. Let $\{x_n\}_n$ be Cauchy in $(X, \tilde{d})$. Given $\epsilon > 0$, let $\delta = \epsilon/(1 + \epsilon)$ and choose $N \in \mathbb{N}$ such that for all $m, n \geq N$ we have

$$\frac{d(x_n, x_m)}{1 + d(x_n, x_m)} = \tilde{d}(x_n, x_m) < \delta = \frac{\epsilon}{1 + \epsilon}.$$

Rearranging and canceling terms, we obtain $d(x_n, x_m) < \epsilon$. Thus, $x_n \rightarrow x$ in (X, d) for some $x \in X$. Now the observation that $\tilde{d} \leq d$ concludes the proof that $x_n \rightarrow x$ in $(X, \tilde{d})$.

We leave the other direction as an exercise.

□